

Knowledge Distillation for Lightweight and Explainable Intrusion Detection in Resource-Constrained Consumer Devices

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Abstract—The growing demand for deploying deep learning (DL) models for cyber-attack detection on consumer devices is constrained by their limited memory and computational resources. High-performing DL models are often too resource-intensive for on-device deployment. Knowledge distillation (KD) offers an alternative by transferring knowledge from a complex teacher model to a smaller, lightweight student model, reducing model size and inference time with minimal accuracy loss. In this paper, a KD-based framework is utilized to design and train both teacher and student models for intrusion detection using the N-BaIoT and CIC-IDS 2023 datasets. The student model achieves comparable performance to the teacher, with accuracies of 99.87% and 98.71% on the utilized datasets. It maintains accurate classification across various attack classes, achieving F1-scores above 0.98, while reducing computational complexity from 263k to 120k trainable parameters, model size from 1.00 MB to 471.67 KB, and inference time from 48 ms to 34 ms. Moreover, SHAP-based explainable AI techniques are employed to interpret both models, demonstrating that the distilled student model retains key feature attributions of the teacher model.

Index Terms—Knowledge distillation, deep learning, consumer electronics, IoT attacks, SHAP, teacher model, student model.

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I. INTRODUCTION

THE RAPID advancement in technology has resulted in the widespread development and adoption of consumer devices [1]. Consumer Internet of Things (IoT) technologies, including smart home systems, wearables, healthcare monitors, and connected appliances, play a key role in bridging the digital and physical worlds [2]. Recently, IoT devices have been transitioning toward the Artificial-Intelligence-of-Things (AIoT), where AI enables intelligent decision-making locally on these devices [3]. However, integrating AI functionality directly onto devices introduces new security vulnerabilities [4]. Hackers can exploit these devices to steal sensitive data, causing security issues for users.

Advanced deep learning (DL) algorithms are essential for detecting evolving threats due to their ability to learn complex patterns and update dynamically. However, their high computational cost limits edge deployment, necessitating lightweight alternatives [5]. To address this challenge, researchers have developed several techniques for building lightweight DL models. A promising solution is Knowledge Distillation (KD), in which a large and complex teacher model is trained on the dataset to achieve high accuracy [6]. A smaller student model then learns from the teacher by mimicking its outputs and refining its parameters using the dataset's labels. This process transfers the knowledge from the teacher to the student, resulting in a lightweight model that retains the same accuracy as the teacher model.

Recent studies have explored various lightweight models for intrusion detection and IoT attacks in consumer electronics. Most of the works [7], [8], [9] utilize DL and ML to reduce computational complexity while maintaining DL model performance. While some of the proposed DL-based approaches [7], [9], [10], [11] exhibit strong performance but most of these require large training datasets and high-specification hardware, which limits their suitability for deployment on resource-constrained edge devices. To address these issues, KD [6], [12], [13], [14] has therefore emerged as an alternative solution for reducing model size while retaining accuracy. These KD-based solutions highlight the potential of distillation methods in intrusion and attack detection. However, most of the studies are not tailored to consumer electronics and edge devices.

The key challenge addressed in this paper is developing a DL-based intrusion detection model that is both lightweight

for resource-constrained consumer devices and explainable for improved interpretability. Conventional DL models generally exhibit high computational complexity involving complex decision-making processes, making them unsuitable for on-device deployment in sensitive IoT environments.

II. THE PROPOSED METHODOLOGY

In this section, the methodology and processes employed in the study are presented. As illustrated in Fig. 1, the overall workflow comprises several main stages: (1) Dataset selection, including N-BaIoT and CIC-IDS 2023; (2) Data Preprocessing, encompassing outlier removal, data balancing, standardization, and splitting; (3) Knowledge Distillation and (4) Evaluation and Detection, in which the trained models are assessed for intrusion detection performance. These stages are discussed in detail in the following subsections.

A. Dataset

This study utilizes two datasets, N-BaIoT [15], [16] and CIC-IDS 2023 [17], [18], to capture a wide range of IoT attacks and everyday consumer electronics usage. N-BaIoT provides authentic traffic data from multiple commercial IoT devices compromised by botnets. CIC-IDS 2023, on the other hand, includes various attack scenarios across an extensive IoT topology with diverse devices acting as both attackers and victims.

1) *Dataset Preprocessing*: The N-BaIoT dataset comprised 7,062,606 rows $\times$ 116 columns, and 11 classes, whereas the CIC-IDS 2023 dataset contained 46,686,579 rows $\times$ 47 columns, and 34 classes. However, in the CIC-IDS 2023 dataset, several classes represented attacks that belonged to the same category, and some classes had very few samples. To address these issues, such classes were merged into their respective attack categories, resulting in a total of 6 aggregated classes. Thus, for the CIC-IDS 2023 dataset, a total of 85,915 samples among 6 classes were utilized for this study. The raw data files were first converted into unified DataFrames for each dataset. Then, a standard scaler was applied to normalize the features, ensuring consistent ranges across all variables and enhancing compatibility for DL models. Furthermore, outliers were removed using IQR-based filtering, and missing values were discarded to ensure quality data input. Only data points that satisfy (1) are retained, which reduces noise without compromising the dataset's structure.

$$Q_1 - 1.5 \times IQR \leq X \leq Q_3 + 1.5 \times IQR \quad (1)$$

Feature selection has been performed using correlation analysis, random forest feature importance, and principal component analysis (PCA). Top-ranking features identified by the aforementioned techniques undergo a majority vote, i.e., a feature was retained if it appeared among the highest-ranked features in at least two out of the three selection methods. A radar (spider) chart, shown in Fig. 2(d), compares the normalized feature importance scores from each technique. This visualization highlights both the overlap and divergence among selected features during the selection process.

B. Knowledge Distillation

The overall KD workflow for training both teacher and student models on the N-BaIoT and CIC-IDS 2023 datasets is shown in Fig. 3. Each model receives the same input samples and produces a set of logits, denoted by $\mathbf{z}_T$ for the teacher and $\mathbf{z}_S$ for the student. A softmax function converts these logits into class probabilities is given in (2):

$$\hat{p}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)}, \quad (2)$$

where $z_i \in \mathbf{z}$ corresponds to the i -th logit, and C is the total number of classes.

The student model learns directly from ground-truth labels by minimizing the following cross-entropy (hard) loss as given in (3).

$$\mathcal{L}_{\text{hard}} = - \sum_{i=1}^C y_i \log(\hat{p}_{S,i}), \quad (3)$$

where y_i is 1 for the correct class and 0 otherwise, and $\hat{p}_{S,i} = \sigma(\mathbf{z}_S)_i$ is the student's softmax output for class i .

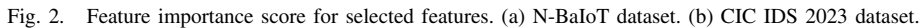
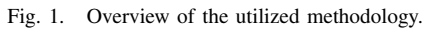
In addition to the hard loss, the student receives guidance from the teacher's softened outputs via the distillation loss $\mathcal{L}_{KD}$. Let $\hat{p}_{T,i} = \sigma(\mathbf{z}_T/T)_i$ and $\hat{p}_{S,i} = \sigma(\mathbf{z}_S/T)_i$ be the teacher's and student's probabilities at temperature T . Mathematically it is given in (4).

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{\text{hard}} + (1 - \alpha) T^2 \text{KL}(\hat{p}_T \| \hat{p}_S), \quad (4)$$

where $\text{KL}(\cdot, \cdot)$ is the Kullback–Leibler divergence [19], and α balances the weight between the hard loss and the teacher–student alignment term. The factor T^2 scales the divergence when $T > 1$.

As shown in Fig. 3, both the teacher and student models take the same inputs from the dataset. The student's logits $\mathbf{z}_S$ produce probabilities $\hat{p}_{S,i}$ used in the hard loss $\mathcal{L}_{\text{hard}}$ and the teacher's logits $\mathbf{z}_T$ generate soft probabilities $\hat{p}_{T,i}$. The student mimics these distributions through the distillation loss $\mathcal{L}_{KD}$. Only the student receives gradients from $\mathcal{L}_{KD}$, creating a smaller model with competitive performance at reduced computational cost. By combining standard cross-entropy (ground-truth labels) with the soft-label guidance from the teacher, the distilled student model preserves much of the teacher's predictive power while operating with lower resource requirements. The utilized KD pipeline is given in Algorithm 1.

1) *The Teacher Model*: The teacher model in KD is typically more complex and larger, allowing it to learn complex representations from the data. In this study, the teacher model's architecture, shown in Fig. 4, comprises multiple convolutional blocks, each containing two 1D convolutional (1D-CNN) layers. The first convolutional block utilizes 32 filters with ReLU activation, while the second has 64 filters. Both use a kernel size of 2×2 and a max-pooling operation with a 2×2 window. Following these convolutional blocks, two LSTM layers with 64 units each form the LSTM block. Finally, two dense layers provide the final classification: the first dense layer has 32 units, and the output layer has as many



2) *The Student Model:* The student model (as shown in Fig. 4) adopts a similar structure but in a lighter configuration. It includes only one 1D-CNN layer in the convolutional block and a single LSTM layer in the LSTM block, along with batch normalization and max-pooling. This design reduces the number of parameters significantly, resulting in a more compact model suitable for resource-constrained edge deployment in consumer electronics.

This section describes the metrics used to evaluate classification performance and model reliability.

2) *Reliability Metrics:* In addition to standard metrics, this paper reports Cohen’s Kappa and Matthews Correlation Coefficient (MCC). Both are particularly useful in scenarios with imbalanced data: Cohen’s Kappa accounts for the possibility of agreement between predicted and true labels, while MCC measures the correlation between observed and predicted classifications, providing a more informative assessment than raw accuracy when class distribution is skewed.

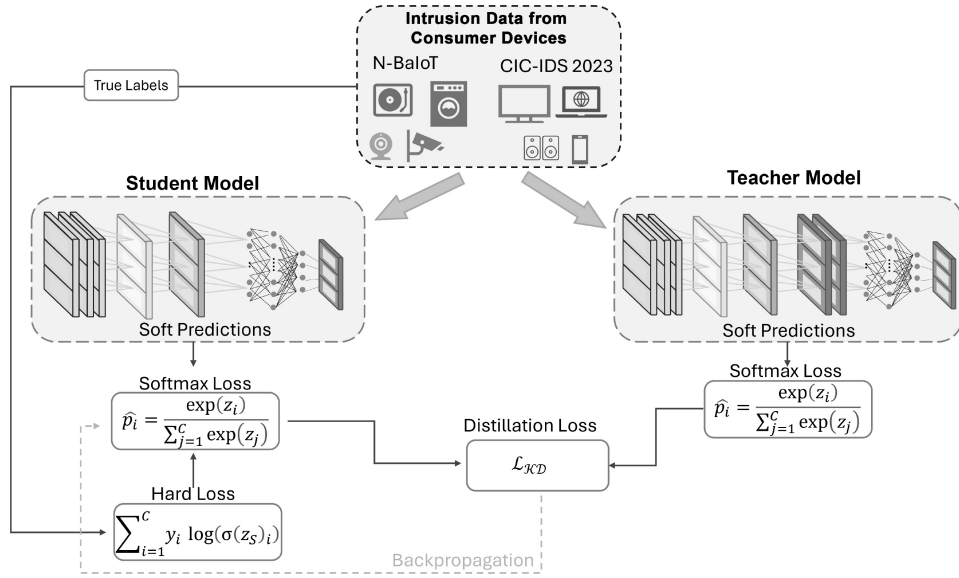


Fig. 3. The knowledge distillation workflow illustrating teacher–student model interaction.

Algorithm 1 The Utilized Knowledge Distillation Pipeline

DATA PREPARATION($\mathcal{D}$)

Clean and remove outliers in $\mathcal{D}$
 Balance classes (under-sampling)
 Standardize features
 $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \leftarrow \text{Split}(\mathcal{D}, \rho)$

TEACHER TRAINING(M_{teach})

$M_{\text{teach}} \leftarrow \arg \min \mathcal{L}_{\text{CE}}(M_{\text{teach}}, \mathcal{D}_{\text{train}})$
return M_{teach}

DISTILLATION($M_{\text{teach}}, M_{\text{stud}}, \alpha, T$)

$\hat{p}_T \leftarrow \sigma(\frac{z_T}{T})$ (teacher probabilities)
 $\hat{p}_S \leftarrow \sigma(\frac{z_S}{T})$ (student probabilities)
 $\mathcal{L}_{KD} = \alpha \mathcal{L}_{\text{hard}} + (1 - \alpha) T^2 \text{KL}(\hat{p}_T \parallel \hat{p}_S)$
 $M_{\text{stud}} \leftarrow \arg \min \mathcal{L}_{KD}(M_{\text{stud}}, \mathcal{D}_{\text{train}})$
return M_{stud}

EVALUATION($M_{\text{teach}}, M_{\text{stud}}, \mathcal{D}_{\text{test}}$)

Evaluate both models on $\mathcal{D}_{\text{test}}$ (ACC, F1, etc.)
return Performance metrics

TABLE I
PERFORMANCE METRICS OF THE TEACHER MODEL ON N-BalIoT DATASET

Class	Precision	Recall	F1-Score	Samples
benign	0.98	0.99	0.99	2037
gafgyt_scan	1.00	1.00	1.00	2034
gafgyt_tcp	1.00	0.99	0.99	2011
gafgyt_udp	0.98	1.00	0.99	2072
gafgyt_combo	0.99	1.00	0.99	2002
gafgyt_junk	0.98	0.99	0.99	1968
mirai_syn	0.99	0.98	0.99	1997
mirai_udp	0.98	0.99	0.98	1943
mirai_ack	0.99	0.99	0.99	1973
mirai_scan	1.00	1.00	1.00	1974
mirai_udpllain	1.00	0.99	0.99	1989
Macro Avg.	0.99	0.99	0.99	22000
Accuracy	99.88%			
Kappa	0.98			
MCC	0.98			

III. RESULTS AND DISCUSSION

In this section, the results are presented for the teacher and student models. Both models are trained on 50 epochs with the Adam optimizer (learning rate 10^{-4}) and a batch size of 32. These parameters are selected via Keras Tuner, which automates hyperparameter optimization. Fig. 5(d) depicts the training curves for both models over epochs.

A. Performance Evaluation

The teacher model, as shown in Table I, achieves an overall accuracy of 99.88%, and its confusion matrix is shown in Fig. 6(a), which shows the numbers of correctly predicted samples. Its macro-average precision, recall, and F1-score values indicate balanced performance across all attack classes, including benign traffic. The results of Cohen’s Kappa (0.98) and MCC (0.98) exhibit an overall best performance between predicted and true labels, highlighting model’s reliability and validates its performance on unseen data. In comparison, the student Model (as given in Table II) shows a minor

3) *Model Complexity*: Model complexity is evaluated based on the following criteria: number of parameters, where fewer parameters reduce computational overhead; memory footprint, as a smaller size enables deployment on resource-constrained devices; and inference time, since faster inference is crucial for real-time environments.

4) *Explainable AI*: DL models are often complex and uninterpretable, making it challenging to understand and interpret their behavior. To address this, SHapley Additive Explanations (SHAP) have been utilized in this study. SHAP facilitates the interpretation of the knowledge transfer process from a larger “teacher” model to a smaller “student” model.

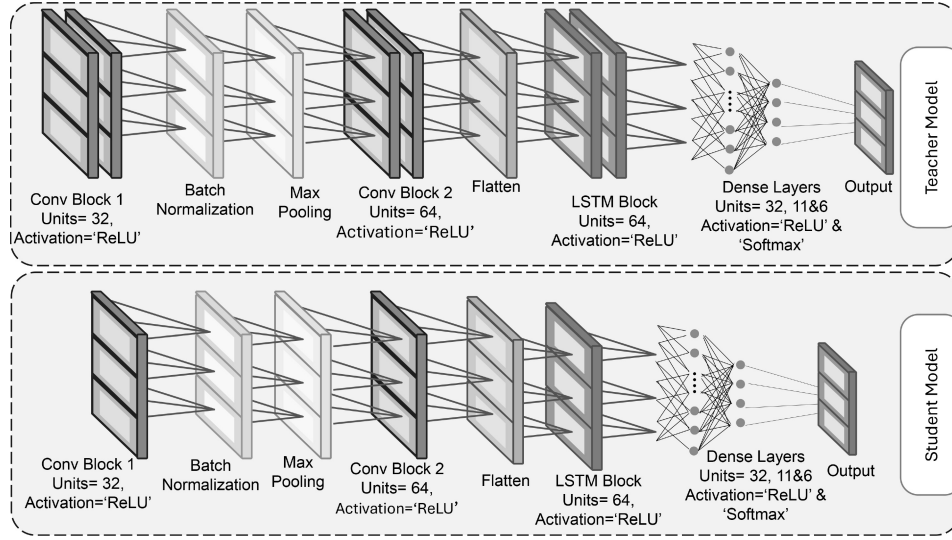


Fig. 4. Deep learning model architecture for teacher and student frameworks.

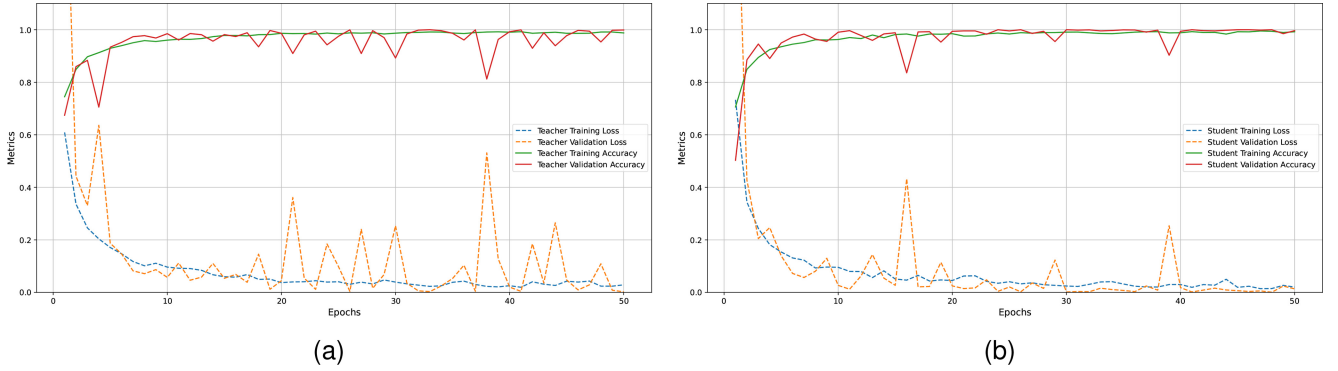


Fig. 5. Training and validation curves for utilized DL models. (a) Teacher Model. (b) Student Model.

 TABLE II
PERFORMANCE METRICS OF THE STUDENT MODEL ON N-BAIoT DATASET

Class	Precision	Recall	F1-Score	Samples
benign	0.97	0.98	0.98	2037
gafgyt_scan	0.99	1.00	0.99	2034
gafgyt_tcp	1.00	0.99	0.99	2011
gafgyt_udp	0.98	1.00	0.99	2072
gafgyt_combo	0.99	1.00	0.99	2002
gafgyt_junk	0.98	0.98	0.98	1968
mirai_syn	0.98	0.98	0.98	1997
mirai_udp	0.99	0.99	0.99	1943
mirai_ack	0.97	0.99	0.98	1973
mirai_scan	1.00	1.00	1.00	1974
mirai_udpplain	1.00	0.98	0.99	1989
Macro Avg.	0.98	0.99	0.99	22000
Accuracy		99.87%		
Kappa		0.97		
MCC		0.97		

 TABLE III
PERFORMANCE METRICS OF THE TEACHER MODEL ON CIC-IDS 2023 DATASET

Class	Precision	Recall	F1-Score	Samples
DDoS	0.98	0.98	0.98	17183
DoS	0.98	0.98	0.98	17183
Benign	0.99	0.99	0.99	17183
Other Attacks	0.99	0.99	0.99	17183
Reconnaissance	0.99	0.99	0.99	17183
Macro Avg.	0.99	0.99	0.99	85915
Accuracy		98.95%		
Kappa		0.98		
MCC		0.98		

difference in accuracy (i.e., 99.87%) compared to the teacher model, confirming that much of the trained teacher model knowledge has been transferred effectively through the KD process. The corresponding confusion matrix is shown in Fig. 6(b).

Whereas, for the CIC IDS 2023 dataset, the teacher model achieves an overall accuracy of 98.95% (as given in Table III)

across five classes: DDoS, DoS, Benign, Other Attacks, and Reconnaissance. Table III indicates better scores for precision, recall, and F1-scores among all classes, with most values exceeding 0.98. Benign and Reconnaissance categories demonstrate especially strong performance, each surpassing 0.99 in precision, recall, and F1-score. Its confusion matrix has been shown in Fig. 6(c). However, the student model (as given in Table IV) achieves an overall accuracy of 98.71% with precision and F1-scores generally ranging from 0.98 to 0.99. Despite a reduced architecture, Cohen's Kappa and MCC values, both at 0.97, further indicate best performance

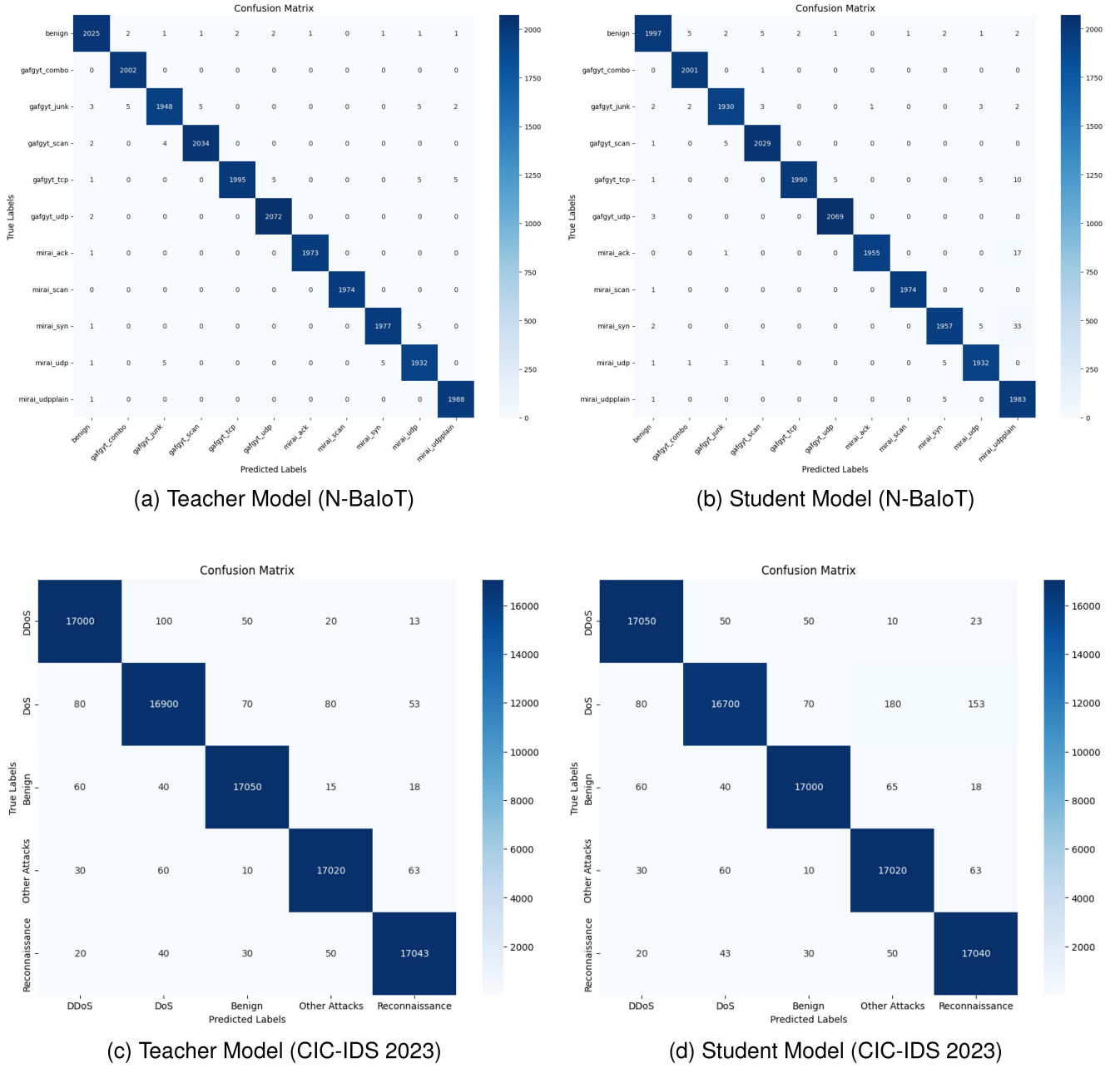


Fig. 6. Confusion matrices of teacher and student models on N-BaIoT and CIC-IDS 2023 test sets.

TABLE IV
PERFORMANCE METRICS OF THE STUDENT MODEL ON CIC-IDS 2023 DATASET

Class	Precision	Recall	F1-Score	Samples
DDoS	0.98	0.99	0.99	17183
DoS	0.98	0.97	0.98	17183
Benign	0.99	0.98	0.99	17183
Other Attacks	0.98	0.99	0.98	17183
Reconnaissance	0.98	0.99	0.98	17183
Macro Avg.	0.99	0.98	0.99	85915
Accuracy		98.71%		
Kappa		0.97		
MCC		0.97		

of model between the predicted and true labels, confirming the model's reliability. Its confusion matrix has been shown in Fig. 6(d).

To demonstrate the statistical reliability of obtained results, confidence intervals (CIs) of the trained models with 95% CIs have been computed. For the N-BaIoT dataset (22,000 test samples), the teacher model achieved a CI of [99.86%, 99.90%], while the student model achieved [99.85%, 99.89%]. On the CIC-IDS 2023 dataset (85,915 test samples), the teacher model yielded a CI of [98.88%, 99.03%], whereas the student model attained [98.63%, 98.79%]. These intervals reflect the reliability due to large sample sizes and support the robustness of reported performance metrics.

Moreover, to assess the model's generalizability, a 5-fold cross-validation on the N-BaIoT and CIC-IDS2023 datasets has been performed. Across all folds, the teacher model achieved an average accuracy of $99.89\% \pm 0.02\% \pm 0.07\%$ on N-BaIoT and $98.94\% \pm 0.10\% \pm 0.21\%$ on CIC-IDS2023;

TABLE V
LAYER-WISE INFERENCE TIMES OF TEACHER AND STUDENT MODELS

Teacher Layers	Time (ms)	Student Layers	Time (ms)
Conv 1	16.23	Conv 1	15.24
Conv 2	8.12	×	—
Batch Norm	2.10	Batch Norm	2.45
Max Pool	2.03	Max Pool	2.35
Conv 3	3.14	Conv 3	3.25
Conv 4	3.65	×	—
Batch Norm	2.04	Batch Norm	1.65
Max Pool	2.01	Max Pool	2.04
LSTM 1	2.65	LSTM 1	3.02
LSTM 2	2.34	×	—
Flatten	1.50	Flatten	1.62
Dense 1	1.26	Dense 1	1.75
Dense 2	1.35	Dense 2	1.02
# Layers	13	# Layers	10
Avg. per Layer	3.72 ms	Avg. per Layer	3.44 ms

TABLE VI
MODEL SIZES AND PARAMETER COUNTS OF TEACHER AND STUDENT MODELS

Teacher Model			Student Model		
Layers	Output	Param.	Layers	Output	Param.
Conv1D	(29,32)	96	Conv1D	(28,32)	128
Conv1D	(28,32)	2,080	-	-	-
BatchNorm	(28,32)	128	BatchNorm	(28,32)	128
MaxPool	(14,32)	0	MaxPool	(14,32)	0
Conv1D	(13,64)	4,160	Conv1D	(12,32)	3,104
Conv1D	(12,64)	8,256	-	-	-
Flatten	(768)	0	Flatten	(384)	0
LSTM	(1,768)	213,248	LSTM	(1,64)	114,944
LSTM	(1,64)	33,024	-	-	-
Dense	(1,32)	2,080	Dense	(1,32)	2,080
Dense	(1,11)	363	Dense	(1,11)	363
Total		263,435	Total		120,747
Size		1.00MB	Size		471.67KB

the student model maintained competitive results of $99.85\% \pm 0.02\% \pm 0.05\%$ and $98.70\% \pm 0.12\% \pm 0.09\%$, respectively. To evaluate robustness against unseen threats, small test set (not utilized in the training folds) has been used. Both models demonstrated an accuracy over (over 98%), with only a marginal performance drop.

B. Model Size

Ensuring a lightweight architecture is important for deploying DL models on consumer electronics. Table VI highlights the parameter counts and layer configurations for both the teacher and student models. Despite sharing similar architectural components, the student model reduces the total number of parameters from approximately 263k to 120k and the model size from 1.00 MB to 471.67 KB. This parameter reduction enables faster inference and lower resource consumption.

C. Inference Time

The teacher model takes a total of 48.42 ms for a single forward pass, which averages to 3.72 ms per layer. Whereas, the student model exhibits a reduced inference times of 34.39 ms with an average of 3.44 ms per layer. Table V presents the inference times for both models, reflecting the impact of model complexity.

TABLE VII
COMPARISON OF BASELINE MODEL (NO KD) AND KD STUDENT MODEL (~ 120K PARAMETERS)

Baseline (No KD)			KD Student (proposed)		
Metric	N-BaIoT	CIC-IDS	Metric	N-BaIoT	CIC-IDS
Accuracy	98.95%	97.98%	Accuracy	99.87%	98.71%
Precision	0.988	0.979	Precision	0.998	0.986
Recall	0.991	0.978	Recall	0.997	0.988
F1-Score	0.989	0.978	F1-Score	0.998	0.987
Inf. Time	36 ms	37 ms	Inf. Time	34 ms	34 ms
Size	468 KB	469 KB	Size	472 KB	471 KB

TABLE VIII
COMPARISON WITH EXISTING STUDIES

Reference	Dataset	Model	Accuracy
Alshehri et al. [7]	N-BaIoT	SkipGateNet	99.91%
Javed et al. [8]	CICIDS-2018	1D CNN + LSTM	99.57
Chen et al. [9]	CICIDS-2017	Deep Forest	99.97
Farhan et al. [11]	CIC-IDS2017	Deep Stack	95.57%
The Proposed Model	N-BaIoT	KD	99.88%, 98.95%
	CIC-IDS 2023		99.87%, 98.71%

D. Comparative Analysis

To further highlight the advantages of utilized KD approach beyond mere complexity reduction, we conducted additional experiments comparing the utilized KD student model to a tuned baseline model of similar size (i.e., approximately 120k parameters). The baseline model was trained from scratch without teacher logits. Both models were assessed using the N-BaIoT and CIC-IDS 2023 datasets, under identical training conditions. Table VII summarizes the key results and shows that the student model outperformed the baseline model. Furthermore, the comparison Table VIII shows that KD yields a compact student network that attains 99.88% and 99.87% accuracy on the N-BaIoT and CIC-IDS 2023 datasets, respectively, overtaking earlier architectures such as SkipGateNet, Deep Forest, and Deep Stack.

E. Computational Complexity on Edge Device

To illustrate real-world feasibility, both the teacher and student models have been deployed on a mobile device having 2 GB of RAM, a Snapdragon680 octa-core processor (max 2.40 GHz), and OS version 1.0.8.0.TGCMIXM. Using a TensorFlow Lite (TFLite) application for on-device inference, it has been observed that the teacher model required an average of 48 ms per inference and occupied 1.00 MB of storage. In contrast, the distilled student model achieved a reduced inference time of 34 ms and consumed a size of only 471.67 KB. This reduction in latency and memory highlights the student model's suitability for resource-constrained scenarios.

F. Explainable AI

In order to provide interpretability for both the teacher and student models, the SHAP technique has been utilized. Fig. 7 shows the resulting SHAP summary plots for the student model on the N-BaIoT dataset. Each horizontal "violin" corresponds to one feature, with the width of the violin indicating

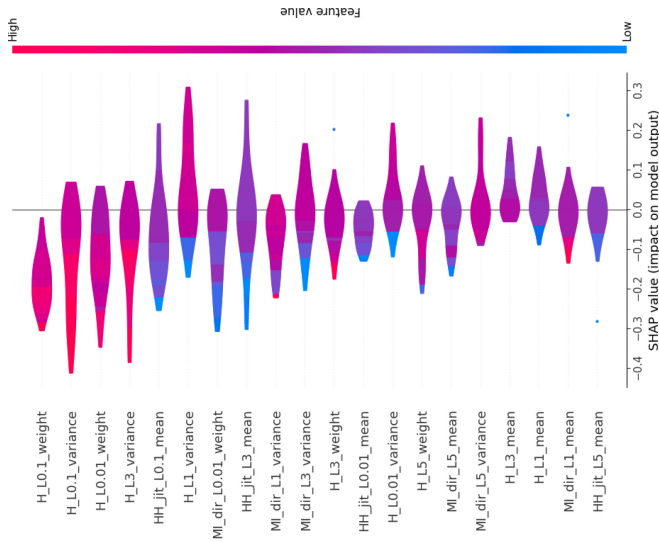


Fig. 7. SHAP summary plot for the **student model**. The color gradient from blue (low) to red (high) reflects each feature's original value range.

how the SHAP values are distributed across all test samples. A positive SHAP value indicates that the feature increases the predicted probability of the target class, whereas a negative value decreases it. The farther from zero, the stronger the influence on the prediction.

IV. CONCLUSION

This study demonstrated and proposed a Knowledge Distillation (KD)-based lightweight and efficient deep learning (DL) model for cyberattack detection in consumer electronics. The distilled student model achieved high performance, with accuracies of 99.87% on the N-BaIoT dataset and 98.71% on the CIC-IDS 2023 dataset, and F1-scores exceeding 0.98 across multiple attack classes. The model significantly reduced the architectural complexity, lowering the number of trainable parameters from approximately 263k to 120k, decreasing model size from 1.00 MB to 471.67 KB, and reducing average inference time from around 48 ms to 34 ms. For interpretability, SHAP-based explainable AI technique was utilized, revealing that the student model preserved key feature attributions learned by the teacher model. This balance of high accuracy, reduced computational cost, and interpretability highlights the effectiveness and scalability of KD for secure and practical intrusion detection in resource-constrained edge environments.

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